

Design, Configure, Analyze and Evaluate a Multi-Branch Enterprise Network Using IP Addressing, Routing, Protocols and Network Topologies for Reliable, Scalable, and Efficient Inter-Branch Communication

1. Problem Statement

A growing organization needs a network to connect its Head Office and two Branch Offices.

The organization has at least **36 end devices** distributed among HR, R&D, Sales and Finance departments.

The network should:

- Connect all three offices.
- Connect different departments.
- Use IPv4 addressing.
- Use subnetting/VLSM.
- Provide suitable default gateways.
- Configure routing between different networks.
- Provide communication between branch offices.
- Use suitable routers and switches.
- Design Star, Mesh, Bus and Hybrid topologies.
- Compare the different topologies.
- Test TCP, UDP, HTTP and DNS.
- Test the network under different conditions.
- Measure network performance.
- Identify possible problems and bottlenecks.
- Select the best topology for the final network.

These requirements are directly given in the assignment problem statement.

2. Objectives

The main objectives of this assignment are:

1. To design a multi-branch enterprise network.
2. To understand IPv4 addressing and subnetting.
3. To assign IP addresses to different departments.
4. To configure routers and switches.
5. To configure routing between different networks.
6. To check communication between different branches.
7. To design Star, Mesh, Bus and Hybrid topologies.
8. To compare the advantages and disadvantages of each topology.
9. To understand TCP, UDP, HTTP and DNS protocols.
10. To test the network under different traffic conditions.
11. To measure latency, throughput, packet loss and delay.
12. To select the best topology for the organization.

3. Course Outcomes

This assignment is related to the following Course Outcomes:

CO2

Configure IP addressing schemes, subnetting, and basic routing techniques using a simulation tool for efficient network design and topology.

CO3

Analyze and compare the performance of TCP, UDP, HTTP and DNS under different network conditions.

CO4

Design network topologies such as Star, Mesh, Bus and Hybrid using networking devices, transmission media and cabling standards.

The assignment is mapped to **L5 – Evaluate** and **L6 – Create**. It is also related to **SDG 9 – Industry, Innovation and Infrastructure**.

4. Requirements

Hardware/Network Devices

The following devices are required:

Device	Quantity
Router	3
Switch	3
PC	36
DNS Server	1
HTTP Server	1
Network cables	As required

Software

Cisco Packet Tracer is used for designing and simulating the network.

5. Organization Structure

The organization is divided into three locations.

Head Office

- HR – 8 PCs
- R&D – 8 PCs

Branch Office 1

- Sales – 10 PCs

Branch Office 2

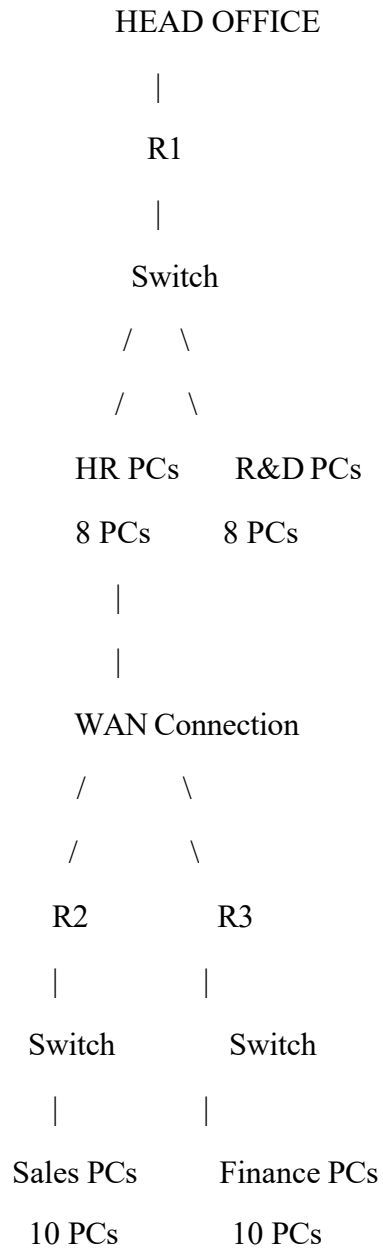
- Finance – 10 PCs

Therefore:

$$8 + 8 + 10 + 10 = 36 \text{ PCs}$$

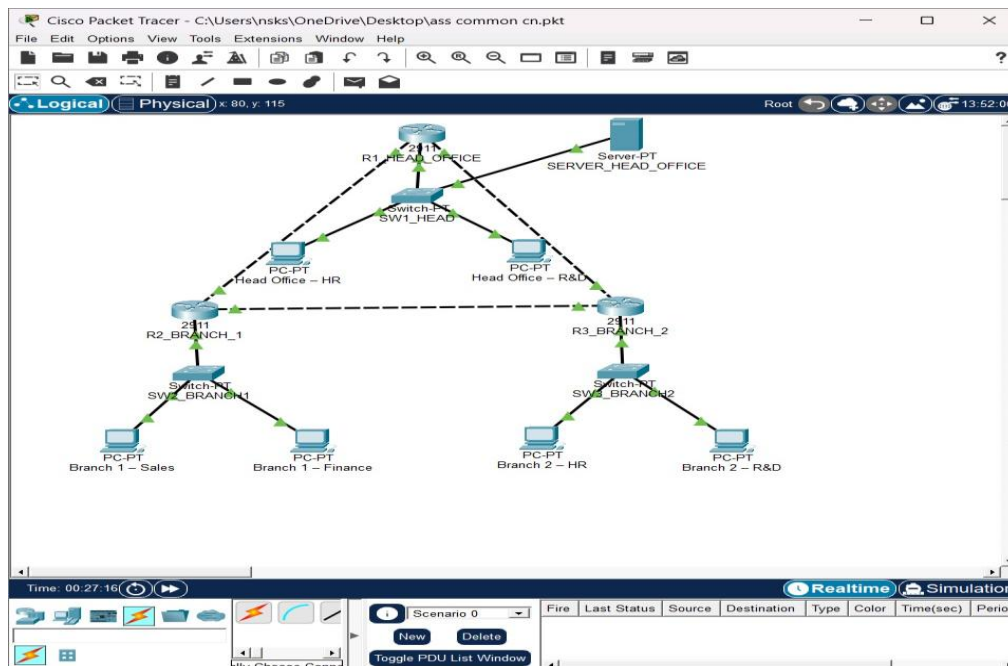
6. Proposed Network Design

We use three routers for connecting the three offices.



The Head Office is connected to both branch offices.

The LAN inside each office can use a **Star topology**, while the complete organization network can be considered a **Hybrid topology**.



7. IP Addressing

For the network, we use:

Network Address: 192.168.10.0/24

We divide this network into smaller networks using subnetting.

Department Networks

Department	Network Address	Subnet Mask	Gateway
HR	192.168.10.0/28	255.255.255.240	192.168.10.1
R&D	192.168.10.16/28	255.255.255.240	192.168.10.17
Sales	192.168.10.32/28	255.255.255.240	192.168.10.33
Finance	192.168.10.48/28	255.255.255.240	192.168.10.49

Each /28 network provides **14 usable IP addresses**, which is suitable for the departments in this example.

8. WAN IP Addressing

The routers are connected using separate small networks.

Connection	Network	Router 1	Router 2/3
R1 – R2	192.168.10.64/30	192.168.10.65	192.168.10.66
R1 – R3	192.168.10.68/30	192.168.10.69	192.168.10.70

The /30 network is useful for router-to-router connections because only two usable IP addresses are required.

9. IP Address Allocation

HR Department

Device	IP Address	Gateway
HR-PC1	192.168.10.2	192.168.10.1
HR-PC2	192.168.10.3	192.168.10.1
HR-PC3	192.168.10.4	192.168.10.1
HR-PC4	192.168.10.5	192.168.10.1
HR-PC5	192.168.10.6	192.168.10.1
HR-PC6	192.168.10.7	192.168.10.1
HR-PC7	192.168.10.8	192.168.10.1
HR-PC8	192.168.10.9	192.168.10.1

R&D Department

Device	IP Address	Gateway
R&D-PC1	192.168.10.18	192.168.10.17
R&D-PC2	192.168.10.19	192.168.10.17
R&D-PC3	192.168.10.20	192.168.10.17
R&D-PC4	192.168.10.21	192.168.10.17
R&D-PC5	192.168.10.22	192.168.10.17
R&D-PC6	192.168.10.23	192.168.10.17
R&D-PC7	192.168.10.24	192.168.10.17

R&D-PC8	192.168.10.25	192.168.10.17
---------	---------------	---------------

Sales Department

Device	IP Address	Gateway
Sales-PC1	192.168.10.34	192.168.10.33
Sales-PC2	192.168.10.35	192.168.10.33
Sales-PC3	192.168.10.36	192.168.10.33
Sales-PC4	192.168.10.37	192.168.10.33
Sales-PC5	192.168.10.38	192.168.10.33
Sales-PC6	192.168.10.39	192.168.10.33
Sales-PC7	192.168.10.40	192.168.10.33
Sales-PC8	192.168.10.41	192.168.10.33
Sales-PC9	192.168.10.42	192.168.10.33
Sales-PC10	192.168.10.43	192.168.10.33

Finance Department

Device	IP Address	Gateway
Finance-PC1	192.168.10.50	192.168.10.49
Finance-PC2	192.168.10.51	192.168.10.49
Finance-PC3	192.168.10.52	192.168.10.49
Finance-PC4	192.168.10.53	192.168.10.49
Finance-PC5	192.168.10.54	192.168.10.49
Finance-PC6	192.168.10.55	192.168.10.49
Finance-PC7	192.168.10.56	192.168.10.49
Finance-PC8	192.168.10.57	192.168.10.49
Finance-PC9	192.168.10.58	192.168.10.49
Finance-PC10	192.168.10.59	192.168.10.49

10. Routing

Routing is required because different departments are using different networks.

We configure the router interfaces with the appropriate IP addresses.

Example Router Configuration

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.240

no shutdown

interface s0/0/0

ip address 192.168.10.65 255.255.255.2

no shutdown

interface s0/0/1

ip address 192.168.10.69 255.255.255.252

no shutdown

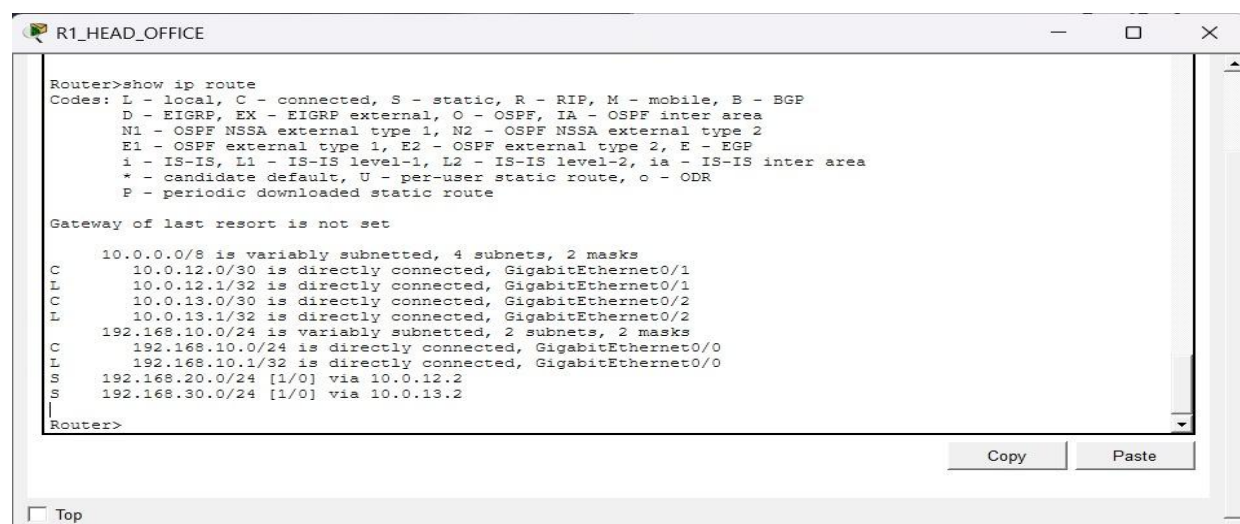
Static routes can then be added to reach the branch networks.

Example:

ip route 192.168.10.32 255.255.255.240 192.168.10.66

ip route 192.168.10.48 255.255.255.240 192.168.10.70

The actual interface names may change depending on the router used in Packet Tracer

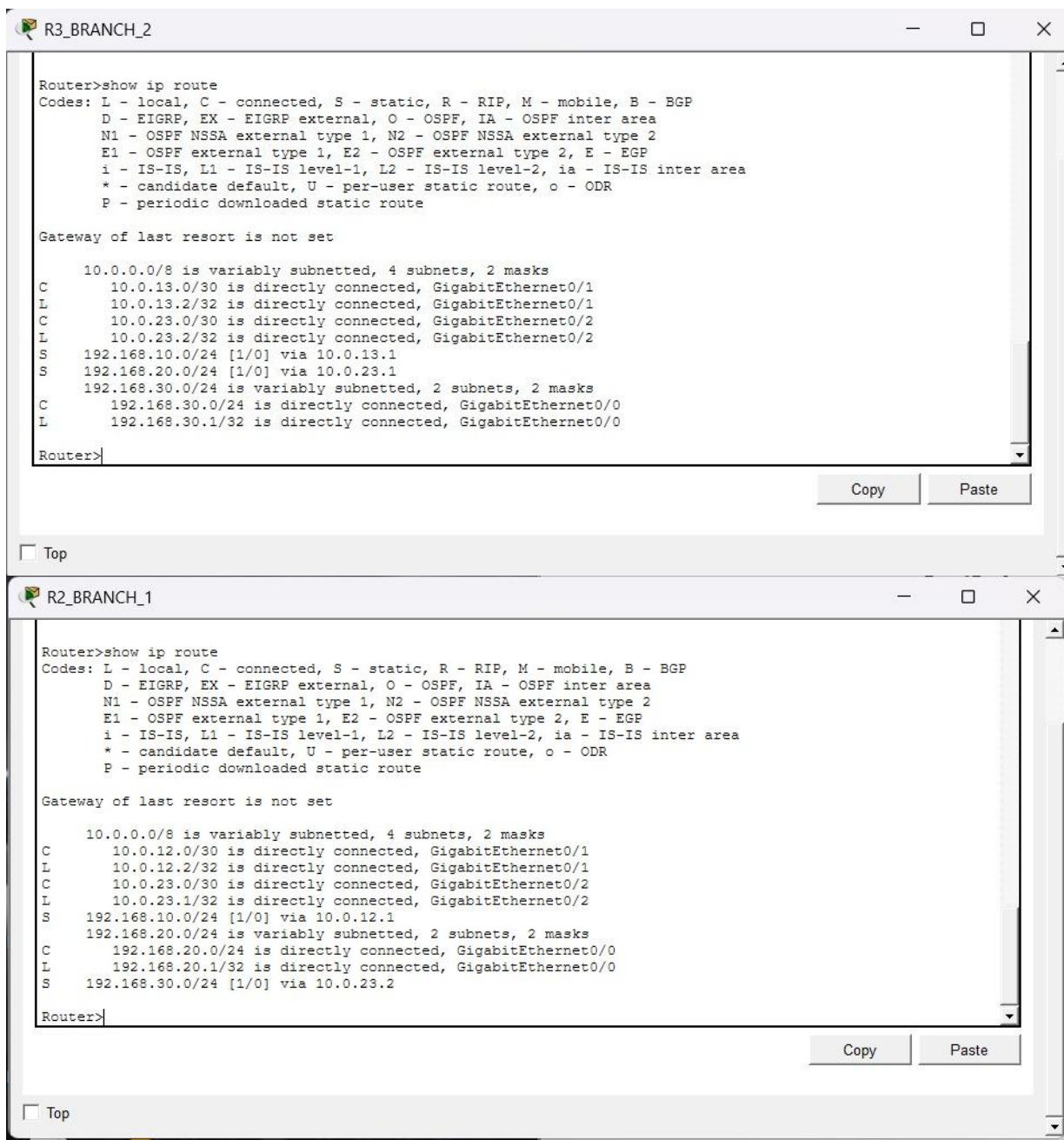


```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, GigabitEthernet0/1
L    10.0.12.1/32 is directly connected, GigabitEthernet0/1
C    10.0.13.0/30 is directly connected, GigabitEthernet0/2
L    10.0.13.1/32 is directly connected, GigabitEthernet0/2
L    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0
S    192.168.20.0/24 [1/0] via 10.0.12.2
S    192.168.30.0/24 [1/0] via 10.0.13.2

Router>
```

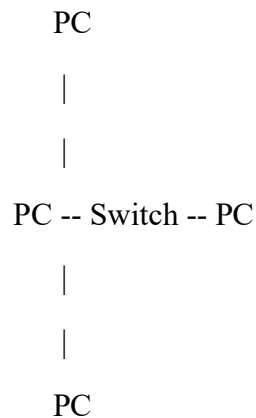



11. Topologies

We studied four different network topologies.

11.1 Star Topology

In Star topology, all computers are connected to a central switch.



Advantages

- Easy to configure.
- Easy to find problems.
- Easy to add new computers.
- Good performance.

Disadvantage:

If the central switch fails, the connected computers cannot communicate.

12. Mesh Topology

In Mesh topology, devices have multiple connections between them.

Router 1

/ \

/ \

Router 2 ----- Router 3

Advantages

- Very reliable.
- Multiple paths are available.
- Good fault tolerance.
- If one connection fails, another path can be used.

Disadvantages

- More cables are required.
- Cost is high.
- Configuration is more difficult.

13. Bus Topology

In Bus topology, all devices share one main cable.

PC ---- PC ---- PC ---- PC

Main Cable

Advantages

- Simple design.
- Less cable is required.
- Low cost.

Disadvantages

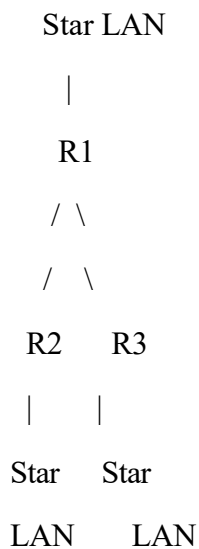
- Main cable failure can affect the network.
- Difficult to troubleshoot.
- Not suitable for large networks.

- Performance can decrease when traffic increases.

14. Hybrid Topology

Hybrid topology combines two or more different topologies.

For our project, we use Star topology inside departments and connect the different offices using routers.



Advantages

- Flexible.
- Easy to expand.
- Good performance.
- Good reliability.
- Suitable for large organizations.

Disadvantage

- More expensive than a simple Star topology.
- Configuration is slightly more difficult.

15. Comparison of Topologies

Parameter	Star	Mesh	Bus	Hybrid
Cost	Medium	High	Low	Medium

Reliability	High	Very High	Low	High
Scalability	High	Medium	Low	High
Fault Tolerance	Medium	Very High	Low	High
Maintenance	Easy	Difficult	Difficult	Medium
Performance	Good	Very Good	Low	Good
Enterprise Use	Good	Good	Not suitable	Best

16. Selected Topology

For our final network, we selected **Hybrid topology**.

The reason is that the organization has multiple offices and departments. A simple Bus topology is not suitable for a large organization.

Mesh topology gives very good reliability but requires more cables and devices.

Star topology is easy to manage, but a single central device failure can affect many devices.

Hybrid topology gives a good balance between **performance, cost, reliability and scalability**.

Therefore, we selected Hybrid topology for the final enterprise network.

17. Network Protocols

We studied four important protocols:

1. TCP
2. UDP
3. HTTP
4. DNS

18. TCP

TCP stands for **Transmission Control Protocol**.

TCP provides reliable communication between two devices.

It checks whether packets are received correctly. If a packet is lost, it can be retransmitted.

Example

When important data is transferred between offices, TCP is useful because reliable delivery is required.

Main points

- Connection-oriented.
- Reliable.
- Error checking.
- Packet retransmission.
- More overhead than UDP.

19. UDP

UDP stands for **User Datagram Protocol**.

UDP does not establish a connection before sending data.

It is faster than TCP because it has less overhead.

Main points

- Connectionless.
- Faster.
- Less overhead.
- Does not guarantee delivery.
- Suitable when speed is more important.

20. HTTP

HTTP stands for **HyperText Transfer Protocol**.

It is used for communication between a web browser and a web server.

For example, a PC can send a request to the HTTP server and receive a webpage.

PC

|

| HTTP Request

↓

Web Server

| HTTP Response



PC

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.013	R3_BRANCH_2	R1_HEAD_OFFICE	TCP
	0.014	R1_HEAD_OFFICE	SW1_HEAD	TCP
	0.015	SW1_HEAD	SERVER_HEAD_OFFICE	TCP
	0.016	SERVER_HEAD_OFFICE	SW1_HEAD	TCP
	0.017	SW1_HEAD	R1_HEAD_OFFICE	TCP
	0.018	R1_HEAD_OFFICE	R3_BRANCH_2	TCP
	0.019	R3_BRANCH_2	SW3_BRANCH2	TCP
	0.020	SW3_BRANCH2	Branch 2 - HR	TCP
	0.020	--	Branch 2 - HR	HTTP
	0.021	Branch 2 - HR	SW3_BRANCH2	TCP
	0.021	--	Branch 2 - HR	HTTP
	0.022	Branch 2 - HR	SW3_BRANCH2	HTTP
	0.022	SW3_BRANCH2	R3_BRANCH_2	TCP
	0.023	SW3_BRANCH2	R3_BRANCH_2	HTTP
	0.023	R3_BRANCH_2	R1_HEAD_OFFICE	TCP
	0.024	R3_BRANCH_2	R1_HEAD_OFFICE	HTTP
	0.024	R1_HEAD_OFFICE	SW1_HEAD	TCP
	0.025	R1_HEAD_OFFICE	SW1_HEAD	HTTP
	0.025	SW1_HEAD	SERVER_HEAD_OFFICE	TCP
	0.026	SW1_HEAD	SERVER_HEAD_OFFICE	HTTP

Reset Simulation ☒ Constant Delay Capturing...

Play Controls

Event List Filters - Visible Events
DNS, HTTP, TCP, UDP

Edit Filters Show All/None

Time: 01:38:10.001 PLAY CONTROLS Event List Realtime Simulation

21. DNS

DNS stands for **Domain Name System**.

DNS converts a domain name into an IP address.

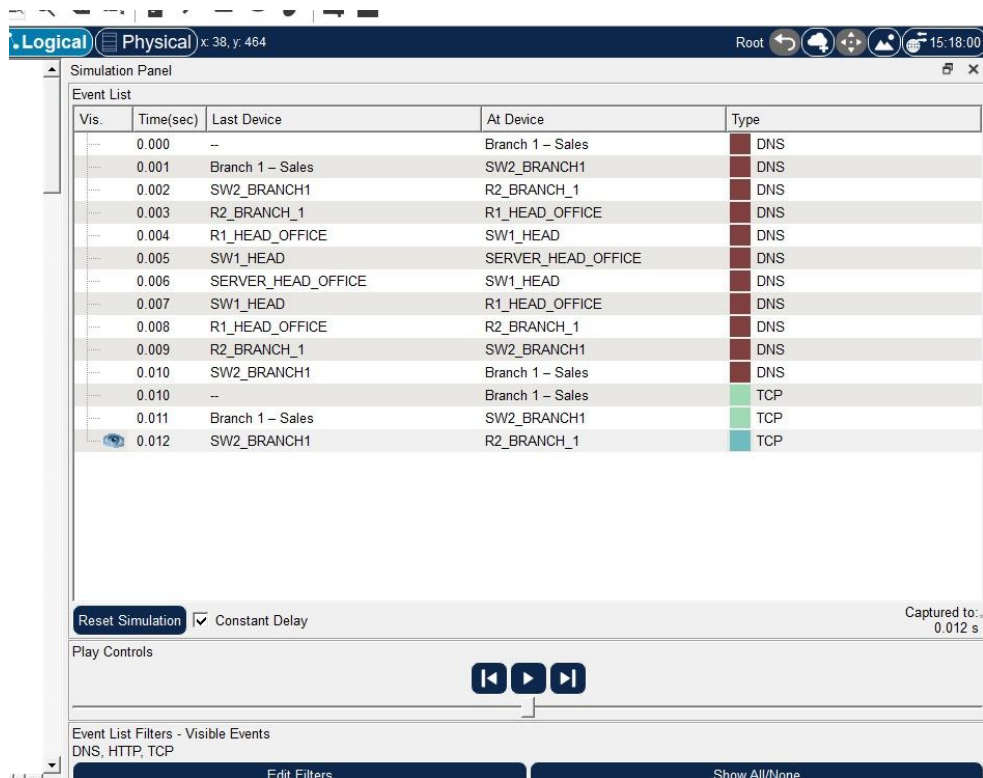
For example:

www.company.com



192.168.10.X

This makes it easier for users because they do not have to remember IP addresses.



22. Protocol Comparison

Feature	TCP	UDP	HTTP	DNS
Type	Transport	Transport	Application	Application
Reliable	Yes	No	Uses TCP	Uses UDP mainly
Speed	Medium	High	Medium	High
Connection	Connection-oriented	Connectionless	Request/Response	Query/Response
Main Use	Reliable data transfer	Fast data transfer	Web communication	Name resolution

23. Simulation Using Cisco Packet Tracer

Cisco Packet Tracer is used to create the network.

Steps followed:

1. Open Cisco Packet Tracer.
2. Add three routers.
3. Add three switches.

4. Add 36 PCs.
5. Add DNS and HTTP servers.
6. Connect PCs to switches.
7. Connect switches to routers.
8. Connect the routers.
9. Assign IP addresses.
10. Configure default gateways.
11. Configure routing.
12. Test the connection using ping.
13. Configure DNS.
14. Configure HTTP.
15. Test TCP and UDP communication.
16. Introduce different network conditions.
17. Record the results.

The assignment requires the simulator to be used for configuration, testing and troubleshooting with supporting evidence.

24. Connectivity Testing

We use the **ping command** to check whether two devices can communicate.

Example:

```
ping 192.168.10.34
```

If communication is successful, we get:

```
Reply from 192.168.10.34
```

```
Reply from 192.168.10.34
```

```
Reply from 192.168.10.34
```

This shows that the source PC can communicate with the destination PC.

25. Test Cases

Test Case	Source	Destination	Expected Result
-----------	--------	-------------	-----------------

TC1	HR-PC1	HR-PC2	Successful
TC2	HR-PC1	R&D-PC1	Successful
TC3	HR-PC1	Sales-PC1	Successful
TC4	HR-PC1	Finance-PC1	Successful
TC5	Sales-PC1	Finance-PC1	Successful
TC6	PC	DNS Server	Successful
TC7	PC	HTTP Server	Successful
TC8	Branch PC	Head Office PC	Successful
TC9	PC	PC after link failure	Failure/Recovery observed

26. Normal Traffic Condition

In the first test, we use normal network traffic.

Only a normal amount of data is sent between devices.

Expected observation

- Communication is stable.
- Packet loss is low.
- Delay is low.
- Throughput is good.
- Services work normally.

27. High Traffic Condition

In the second test, more traffic is generated in the network.

This helps us understand how the network behaves when many devices communicate at the same time.

Expected observation

- Network delay increases.
- Latency increases.
- Throughput may decrease.
- Packet loss may increase.

- Network becomes more busy.

28. Link Failure Condition

In the third test, one network link is disconnected.

For example, the connection between R1 and R2 can be disabled.

Expected observation

- Communication through that path is interrupted.
- Packets may be lost.
- Some services may stop working.
- Recovery can be checked if an alternative route is available.

29. Performance Parameters

The following parameters are used to evaluate the network.

Latency

Latency is the time taken for data to travel from source to destination.

Lower latency is better.

Throughput

Throughput is the amount of data successfully transferred in a given time.

Higher throughput is better.

Packet Loss

Packet loss shows how many packets did not reach the destination.

Lower packet loss is better.

Delay

Delay is the time taken during data transmission.

Jitter

Jitter means variation in packet delay.

Successful Delivery Rate

It shows the percentage of packets that successfully reach the destination.

Recovery Time

It shows how much time the network takes to work again after a failure.

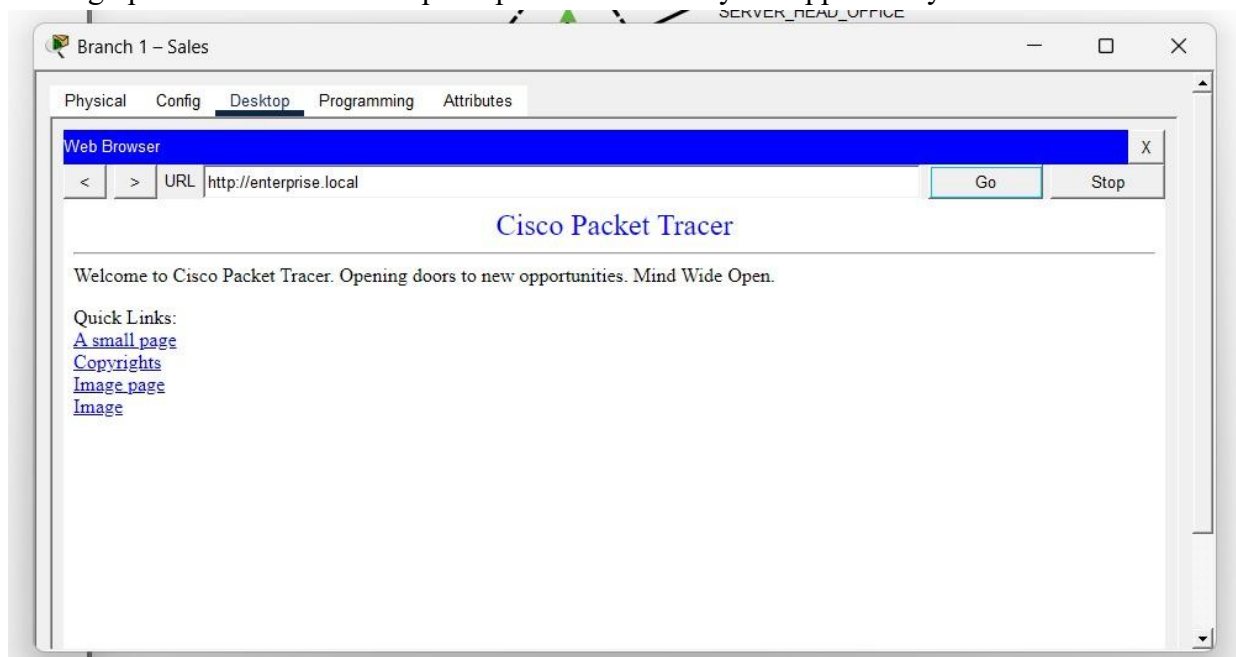
These are the performance measures listed in the assignment requirements.

30. Performance Results

The following table can be filled using the actual values obtained from our Packet Tracer simulation.

Network Condition	Latency	Throughput	Packet Loss	Delay	Recovery Time
Normal Traffic	___ms	___Mbps	___%	___ms	—
High Traffic	___ms	___Mbps	___%	___ms	—
Link Failure	___ms	___Mbps	___%	___ms	___sec

Note: Actual values should be entered from the simulation screenshots/results rather than making up values. The rubric expects performance analysis supported by evidence.



31. Results and Discussion

After configuring the network, we check communication between different departments.

The ping test shows whether devices in different networks can communicate.

The routing configuration allows packets to travel from the Head Office to the Branch Offices.

During normal traffic, the network performs properly.

When traffic increases, the network may experience more delay and packet loss.

When a link failure occurs, communication through that link is affected. If an alternative path is available, communication can be restored.

From the topology comparison, Hybrid topology is more suitable for this organization because it provides good scalability and reliability.

32. Bottlenecks

A bottleneck is a part of the network that slows down communication.

Possible bottlenecks in our network are:

1. Heavy traffic on a WAN link.
2. Failure of a central switch.
3. Limited bandwidth.

4. Too many devices using the same connection.
5. Router processing delay.
6. Network congestion.

To reduce these problems, we can increase bandwidth, use additional links and improve routing.

33. Scalability

The proposed network can be expanded in the future.

For example:

- New PCs can be added.
- New departments can be added.
- New branches can be connected.
- More switches can be installed.
- More IP subnets can be created.

Therefore, the network is suitable for a growing organization.

34. Reliability

The network should continue working even when some problems occur.

Using multiple routers and suitable network connections improves reliability.

Hybrid topology also gives better flexibility because different topology types can be used for different parts of the network.

35. Cost Analysis

Different topologies have different costs.

Bus topology has a low cost but is not suitable for a large enterprise.

Mesh topology provides high reliability but requires more cables and devices, so the cost is high.

Star topology has a medium cost and is easy to manage.

Hybrid topology has a medium to high cost, but it provides better scalability and flexibility.

For this organization, the additional cost of Hybrid topology is acceptable because reliability and future expansion are important.

36. Engineering Decisions

The main decisions made in this project are:

IP Addressing

We selected a private IPv4 address range and divided it into smaller subnets.

Subnetting

Subnetting is used to separate departments and manage IP addresses easily.

Routing

Routing is configured so that different branches can communicate.

Topology

Hybrid topology is selected because it is suitable for a growing organization.

Cabling

Ethernet cables are used for PC-to-switch and switch-to-router connections where suitable.

Protocols

TCP, UDP, HTTP and DNS are selected for studying different types of network communication.

37. Advantages of the Proposed Network

The proposed network provides:

- Easy communication between branches.
- Separate networks for departments.
- Better IP address management.
- Easy troubleshooting.
- Good scalability.
- Good performance.
- Support for common network protocols.
- Better reliability.
- Easy future expansion.

38. Limitations

There are some limitations in our project.

1. The network is simulated using Packet Tracer.
2. It is not a real physical network.
3. Actual network performance may be different.
4. Real hardware failures cannot be fully simulated.
5. Traffic generation is limited by the simulator.
6. Actual bandwidth and latency may differ from the simulation.

39. SDG 9 – Industry, Innovation and Infrastructure

This project is related to **SDG 9** because it focuses on creating reliable and scalable network infrastructure.

A good computer network helps organizations:

- Communicate faster.
- Share resources.
- Connect different offices.
- Support digital services.
- Improve business operations.
- Expand their infrastructure.

Therefore, this project supports the idea of developing reliable digital infrastructure.

40. Conclusion

In this assignment, we designed a multi-branch enterprise network for an organization with a Head Office and two Branch Offices.

We used IPv4 addressing and subnetting to divide the network into different departments. We assigned IP addresses and default gateways to the devices.

We also configured routers and routing so that different departments and branches can communicate with each other.

Star, Mesh, Bus and Hybrid topologies were studied and compared. Based on cost, reliability, scalability and fault tolerance, we selected **Hybrid topology** for the final network.

We also studied TCP, UDP, HTTP and DNS protocols. The network was tested under normal traffic, high traffic and link failure conditions.

From the analysis, we found that network performance changes when traffic increases or when a link fails. Parameters such as latency, throughput, packet loss, delay and recovery time help us understand the network performance.

Overall, this project helped us understand how a real enterprise network can be designed, configured and tested using Cisco Packet Tracer.

41. Future Improvements

In the future, the network can be improved by:

1. Adding more redundant links.
2. Using dynamic routing protocols.
3. Increasing network bandwidth.
4. Adding firewall protection.
5. Adding network monitoring.
6. Adding more servers.
7. Using VLANs for better department separation.
8. Adding wireless networks.
9. Using IPv6 along with IPv4.
10. Adding backup network devices.

42. Individual Contribution

I worked on the IP addressing and subnetting of the enterprise network using VLSM. I calculated the network addresses and subnet masks for the different network segments. I identified the usable host ranges and broadcast addresses for each subnet. I assigned suitable default gateways to the network segments. I configured the router interfaces with the required IP addresses. I assigned IP addresses and default gateways to the end devices. I configured routing between the Head Office and Branch Offices for inter-network communication. I verified the router configurations using appropriate commands. I tested connectivity between different network segments using ping. I also helped identify and resolve connectivity issues during network implementation and verification.

43. Individual Reflection

Reflection – Student 1

During this assignment, I learned how to design and configure a computer network using Cisco Packet Tracer. Before doing this project, I knew the basic concepts of IP addresses and routers, but I understood them better by actually creating the network.

The most challenging part for me was IP addressing and subnetting. At first, it was confusing to decide which IP address should be given to each department. After dividing the main network into smaller subnetworks, it became easier to assign IP addresses and gateways.

I also learned how routers help different networks communicate with each other. The ping test helped me understand whether the configuration was correct. When the ping was unsuccessful, I checked the IP address, subnet mask, gateway and router configuration to find the problem.

Another useful part was learning about TCP, UDP, HTTP and DNS. I understood that different protocols are used for different purposes. TCP is useful when reliable communication is required, while UDP is faster and has less overhead.

The topology comparison was also interesting. I learned that every topology has its own advantages and disadvantages. Finally, we selected Hybrid topology because it is suitable for a large organization and can be expanded easily.

This assignment improved my practical knowledge of computer networks. I also learned how important testing and troubleshooting are when designing a network.

44. References

1. CSA0704 Computer Networks Assignment Specification.
2. Cisco Packet Tracer.

3. Computer Networks course notes.
4. Networking laboratory materials.
5. Relevant computer networking textbooks.

45. Final Deliverables

According to the assignment, the final submission should contain:

No. Deliverable

- 1 Technical Report
- 2 Network Topology Diagram
- 3 IP Addressing and Subnetting Design
- 4 Configuration Screenshots
- 5 Packet Tracer Project File
- 6 TCP, UDP, HTTP and DNS Testing Evidence
- 7 Performance Tables and Graphs
- 8 Engineering Analysis
- 9 Individual Reflection
- 10 Presentation/Demonstration

22843 / 10,000 chars - 3413 Words

Tx



Check Plagiarism via URL

Exclude URLs



Check Plagiarism →

Check Grammar

✓ PCI & NIST Compliant



10% Plagiarized

● Exact Match

10%

● Partial Match

0%



90% Unique

Original Content

Plagiarism Checker

Remove Plagiarism

Check Grammar

AI Detector

Go Pro

Source Breakdown

10+ Sources Found

Document Preview

Words 3413 | Characters 22843